

Phase 3 - Alarm Configuration

Step 3.1 — Create Metric Filter



Prerequisite: You need at least one `ERROR` log entry before creating the metric filter. If you haven't triggered an error yet, you may need to temporarily break something to generate an `ERROR` log.

Console Method

1. **CloudWatch Console** → **Log groups**
2. Click on `/aws/ec2/lab-rds-app`
3. Click **Metric filters** tab → **Create metric filter**
4. **Filter pattern:** `ERROR`
5. Click **Test pattern** to verify it matches your error logs
6. Click **Next**
7. Configure:
 - **Filter name:** `lab-db-error-filter`
 - **Metric namespace:** `Lab/Application`
 - **Metric name:** `DBConnectionErrors`
 - **Metric value:** `1`
8. Click **Create metric filter**

CLI Method

```
aws logs put-metric-filter \  
  --log-group-name /aws/ec2/lab-rds-app \  
  --filter-name lab-db-error-filter \  
  --filter-pattern "ERROR" \  

```

```
--metric-transformations \  
    metricName=DBConnectionErrors,metricNamespace=Lab/Applica  
tion,metricValue=1
```

Step 3.2 — Create CloudWatch Alarm

Console Method

1. **CloudWatch Console** → **Alarms** → **Create alarm**
2. Click **Select metric**
3. Navigate to: **Lab** → **Application** → **DBConnectionErrors**
4. Select the metric and click **Select metric**
5. Configure:
 - **Statistic:** Sum
 - **Period:** 1 minute
 - **Threshold type:** Static
 - **Condition:** Greater than or equal to
 - **Threshold value:** 1
6. **Alarm name:** lab-db-connection-failure-alarm
7. **Description:** "Triggers when DB connection errors are detected"
8. (Optional) Add SNS notification for email alerts
9. Click **Create alarm**

CLI Method

```
aws cloudwatch put-metric-alarm \  
    --alarm-name lab-db-connection-failure-alarm \  
    --alarm-description "Triggers when DB connection errors exc  
eed threshold" \  
    --metric-name DBConnectionErrors \  
    --metric-value 1
```

```
--namespace Lab/Application \  
--statistic Sum \  
--period 60 \  
--threshold 1 \  
--comparison-operator GreaterThanOrEqualToThreshold \  
--evaluation-periods 1 \  
--treat-missing-data notBreaching
```

Step 3.3 — Verify Alarm Exists

```
aws cloudwatch describe-alarms \  
  --alarm-name-prefix lab-db-connection
```



Expected: Alarm appears with `StateValue: "OK"` (no errors yet).

Common Issues — Phase 3

▼ Issue: Can't create metric filter (no matching logs)

Symptom: "Test pattern" shows no matches

Root Cause: No `ERROR` logs exist yet

Fix Options:

1. Temporarily trigger an error (stop RDS, block SG) to generate ERROR logs
2. Manually add a test ERROR line: `echo "ERROR: Test" | sudo tee -a /var/log/rdsapp/app.log`
3. Wait for log to sync to CloudWatch (1-2 minutes)

▼ Issue: Metric namespace doesn't appear

Symptom: Can't find `Lab/Application` in CloudWatch metrics

Root Cause: No ERROR logs have been generated since the metric filter was created

Fix: The metric only appears after at least one ERROR log matches the filter.
Trigger an error first.

▼ Issue: Alarm never fires during incident

Possible Causes:

1. App not logging with `ERROR` keyword
2. Metric filter pattern doesn't match log format
3. Alarm threshold too high
4. Wrong metric namespace selected

Diagnostic:

```
# Check if ERROR logs exist
aws logs filter-log-events \
  --log-group-name /aws/ec2/lab-rds-app \
  --filter-pattern "ERROR" \
  --limit 5
```

Alarm State Reference

State	Meaning	Action
<code>OK</code>	No errors detected	Normal operation
<code>ALARM</code>	Threshold exceeded	Incident detected!
<code>INSUFFICIENT_DATA</code>	No metric data available	Check if metric filter is working

Phase 3 Checklist

- ☐ Metric filter `lab-db-error-filter` created
- ☐ Filter pattern is `ERROR`
- ☐ Metric namespace is `Lab/Application`
- ☐ Metric name is `DBConnectionErrors`
- ☐ Alarm `lab-db-connection-failure-alarm` created

- ☐ Alarm threshold is 1 (or higher if desired)
- ☐ Alarm shows OK state initially